

A MINI PROJECT REPORT
ON
LORA BASED MULTIHOP PREDECTIVE ENERGY DEMAND
MODELLING FOR WIRELESS COMMUNICATION NETWORKS

Submitted for partial fulfillment of the requirements for the
Award of the degree
of
BACHELOR OF TECHNOLOGY
In
ELECTRONICS AND COMMUNICATION ENGINEERING
By

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DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING
BALAJI INSTITUTE OF TECHNOLOGY AND SCIENCE

Accredited by NBA, NAAC A+ & ISO 9001:2015 Certified Institution
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CERTIFICATE

This is to certify that this Project Work entitled "**LORA BASED MULTI HOP PREDICTIVE ENERGY DEMAND MODELLING FOR WIRELESS COMMUNICATION NETWORKS**" is a bonafide work carried out by **Neela Varshika (23C3A04C4)**, **Vaka ravivardhan (23C31A04H1)**, **Sriramula Sharavana (23C31A04G0)**, **Sanga Ravalika (23C31A04F4)**, **Pulishetti Srinivas (23C31A04E3)** in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology** from **JNTUH, Hyderabad** during the period 2025-26 under our guidance and supervision.

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DECLARATION

We, **Neela Varshika (23C3A04C4), Vaka ravivardhan (23C31A04H1), Sriramula Sharavana (23C31A04G0), Sanga Ravalika (23C31A04F4), Pulishetti Srinivas (23C31A04E3)** hereby declare that the project entitled **“LORA BASED MULTIHOP PREDECTIVE ENERGY DEMAND MODELLING FOR WIRELESS COMMUNICATION NETWORKS”**, submitted in the partial fulfillment of the requirements for the award of B.Tech in Electronics and Communication Engineering to Balaji Institute of Technology & Science, Narsampet, affiliated to JNTUH, Hyderabad is a authentic work and has not been submitted to any other university or institution for award of the degree.

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ABSTRACT

A LoRa-based multi-hop wireless communication network architecture is designed to facilitate the efficient collection and transmission of IoT sensor data, combined with external environmental factors, to enable accurate predictive modeling of energy demand. The multi-hop communication paradigm overcomes the inherent range limitations of single-hop LoRa technology by allowing sensor nodes to forward data through intermediary nodes, thereby expanding network coverage and improving data reliability in geographically dispersed sensor deployments. The network architecture incorporates dynamic routing algorithms to optimize data paths and reduce packet loss, ensuring the integrity of transmitted data critical for energy demand analysis. Comprehensive exploratory data analysis (EDA) is conducted on the collected dataset comprising sensor measurements such as voltage, current, temperature, humidity, and external variables including weather conditions and temporal features. Statistical analysis and visualization techniques such as correlation heatmaps, feature distribution plots, and time series decomposition reveal underlying patterns and feature interactions relevant to energy consumption behaviors. Feature selection based on EDA outcomes enhances model efficiency and reduces computational overhead by eliminating redundant or less significant variables. For predictive classification of energy demand categories, a Support Vector Classifier (SVC) is initially employed due to its capability to handle high-dimensional feature spaces and establish clear decision boundaries. However, to address potential limitations of SVC in capturing complex non-linear relationships and to improve robustness against overfitting, a Random Forest Classifier (RFC) is implemented as the proposed model. RFC's ensemble learning mechanism, which aggregates decisions from multiple decision trees trained on random subsets of data and features, enhances prediction stability and accuracy. Model training and validation employ stratified cross-validation techniques to ensure unbiased performance assessment. Experimental results demonstrate that RFC consistently outperforms SVC across multiple evaluation metrics including accuracy, precision, recall, and F1-score. The RFC model exhibits superior generalization on unseen data, providing more reliable predictions of energy demand fluctuations.